Vaidya Setu Hosted MVP Requirements

Product requirements for public demonstration and private consultation review

Document field	Value
Product owner	Akshata Anil Urs
Updated	8 November 2026
PRD version	3.0
Web application	0.3.1 web
Stage	Independent portfolio MVP built with AI assisted coding
Live website	Open Vaidya Setu

Vaidya Setu turns a completed consultation recording or pasted conversation into an editable medication draft. A reviewer checks the source, saves corrections and approves one exact version. The app then supports Kannada, Hindi or English script review and a WAV voice preview. An evaluation archive retains the original draft separately from corrections and source verified reference answers.

The hosted MVP offers a public fictional preview. Uploads, Sarvam calls, saved cases and recordings require ChatGPT sign in and are scoped to the signed in account. The public preview makes no model request or archive write. This product assists review of stated instructions; it does not independently prescribe, issue a signed prescription or establish clinical safety.

1 Problem and intended outcome

Patients and caregivers may need understandable, replayable instructions after a consultation. A language or terminology mismatch can complicate that task. A conversation to audio pipeline can introduce additional errors in drug names, strength, amount per dose, frequency, duration and timing.

The MVP makes these transformations inspectable. Its intended outcome is a source checked medication explanation with a traceable review history. Whether this improves patient comprehension or reduces provider effort remains a research question.

2 Users and product decisions

User	Task	Current support
Recruiter or portfolio visitor	Understand the project without installation	Public fictional preview, research and explanation pages
Reviewer or demo operator	Check and correct medication facts	Source playback, transcript, editable records and saved revisions
Bilingual reviewer	Check translated and spoken instructions	Script editing, back translation, warnings and audio preview
Product evaluator	Distinguish extraction errors from missing source facts	Original draft, field annotations and archived outcomes
Patient or caregiver	Replay instructions in a preferred language	Reviewed audio export; patient testing remains outstanding

These are design roles, not findings from completed interviews. A typed reviewer name does not authenticate a clinician. Sign in establishes an account identity, not professional qualifications.

The product starts with completed recordings so inputs and outputs can be inspected. It separates strength from dose, and time of day from food timing. It builds patient instructions from the approved table. Inferred route suggestions remain separate from spoken facts and extraction scores.

3 Market evidence

Public teleconsultation delivery

MoHFW reported **50,00,92,730 cumulative eSanjeevani consultations** at the milestone reached on **15 September 2026**. Its 6 October release reported approximately **2.5 to 3 lakh daily consultations**, **2,41,923 registered providers** and availability in **15 languages**. These are programme counts, not unique patients or a paid market estimate. MoHFW source

A descriptive study of C DAC programme data reports **more than 163 million consultations during January 2020 to September 2023**, with **over 93%** delivered through the assisted AB HWC model. This historical share should not be presented as the current national or private market mix. Programme study

Product implication to validate: discovery should include the health worker or clinician in an assisted consultation, alongside patients using direct access services. Large utilisation does not establish demand for this particular product.

Digital access and private platforms

MoSPI's January to March 2025 survey of **34,950 households** reported smartphone possession in **85.5% of households** and internet access within the premises in **86.3%**. Household access does not establish individual ownership, upload skills, WhatsApp use or preference for audio. MoSPI source

Apollo's Health of the Nation 2026 report attributes **7.99 million digital consultations in calendar 2025** to Apollo 24/7. This is a company reported platform measure, not a national market total. Practo reports **INR 234 crore FY25 revenue** across its business; this cannot be treated as standalone teleconsultation revenue. Apollo report Practo disclosure

Product implication to validate: test the workflow with potential provider users and patients. Measure actual review effort, comprehension and cost. No TAM, price, market share, revenue forecast or partnership is assumed. The project package includes a fuller market research appendix with dates, source links and discovery questions.

4 Scope and current requirements

ID	Requirement and implemented behaviour	Boundary
F01	Public browsing and varied fictional previews	Preview changes remain in the browser tab and disappear on refresh
F02	ChatGPT sign in for uploads, model requests and archives	Each operation must authorize the account; no owner invitation is needed once the site is public
F03	Completed PCM WAV up to 10 minutes and 30 MB, or pasted text	Pasted text is labelled as bypassing STT
F04	Short file REST STT and longer file batch STT	Batch results are checked while the page is open; pending jobs can be resumed
F05	Up to 12 medication records with evidence and review notes	Unknown facts stay blank; model accuracy is unestablished
F06	Saved corrections and version bound simulated approval	Changed facts or language invalidate dependent outputs
F07	Explicit route plus separate form based suggestions	Suggested routes require confirmation and never become spoken evidence

ID	Requirement and implemented behaviour	Boundary
F08	Source annotations and original draft evaluation	A missing or unverified reference is not a pass
F09	Reviewed Kannada, Hindi or English script and WAV preview	Actual bilingual quality and pronunciation require assessment
F10	Private archive, revision events, case JSON export and deletion	Latest 200 cases and 100 events per case are displayed; export uses the same event limit

The hosted edition does not implement live interpretation, automatic WhatsApp delivery, signed prescriptions, EMR integration, validated FHIR export, interaction or clinical dose checking, a background processing scheduler, FFmpeg OGG conversion or the Python folder benchmark. The earlier Python app remains a separate implementation.

5 User journeys and acceptance criteria

Public visitor

A visitor opens the Live demo link from GitHub, views the research and opens a prepared fictional example without signing in. They can edit its table, confirm a route suggestion and simulate review or field evaluation. The interface labels the data as prepared and the history as local. No anonymous upload, private archive access or billable API operation is allowed.

Signed in consultation review

1. The operator enters a Sarvam key and may test it with a small billable chat request. Entry alone is not verification of a working key or access to every service.
2. The operator uses an alias, chooses the confirmed output language and acknowledges permission for upload, external processing and private archiving.
3. The input is saved. STT and extraction outcomes, including failures, remain visible. The original transcript and accepted original extraction are retained.
4. The reviewer checks the recording, evidence and medication fields. Saving creates a revision event. Clear or new session resets the active UI without deleting archived cases.
5. Required gaps, unclear actions and unresolved route confirmation block approval. A verified reference that contradicts the current plan also blocks or revokes approval.
6. Translation uses approved medication facts. The reviewer saves script edits, runs back translation and records bilingual review before generating speech.
7. The reviewer listens, records that check and downloads WAV. Export does not claim that a WhatsApp message was sent.

Changing medicine facts, route decisions or language clears plan approval and dependent script or audio. Changing a translated script requires its review again. An extraction retry is bounded; explicitly truncated responses are not accepted as medication drafts.

6 Route suggestions and provenance

Source information	Behaviour	Recorded meaning
Extracted explicit route	Preserve it for source review	Extracted value, not automatically verified
Plain tablet, capsule or syrup with no route	Suggest oral by mouth and require confirmation	Form suggestion, separate from the original blank route
Plain cream or ointment with no route	Suggest topical on skin and require confirmation	Form suggestion; application site needs review
Specialised form, drops, injection or unsupported form	Leave generic suggestion unknown	Qualified clarification is required

Tablets and creams can have specialised routes. The app excludes forms such as sublingual or vaginal products from generic mapping. It does not consult a validated drug formulation database. A confirmed suggestion may inform the reviewed explanation, but it cannot count as a correctly extracted doctor statement. MedlinePlus examples FDA route definitions

7 Evaluation archive and data model

The archive retains the input, transcript, original draft, current reviewed table, route decisions, reference annotations, reviewer acknowledgements, processing metadata and generated audio. Review events record saved revisions and relevant snapshots; ordinary edits do not overwrite the original baseline. Deleting a case removes its website records and files.

References attach to medicine record IDs. Original rows are retained for comparison. Reviewers add a row for a missed medicine and mark final plan membership to identify extra or historical medicines. Corrections and reference verification are separate actions.

Each of the ten evaluated fields is annotated as explicitly stated, not mentioned, unclear or unreviewed. The fields are medicine name, action, form, strength, amount per dose, route, frequency, time of day, food timing and duration.

Outcome	Definition
Correct stated field	Original value matches a verified stated reference after conservative text normalization
Incorrect	Original value differs from the stated reference
Missed field	Reference states a value but the original is blank
Correctly blank	Reference confirms not mentioned and the original is blank
Unsupported addition	Reference confirms not mentioned but the original supplies a value
Unclear or unreviewed	Excluded from evaluated field counts
Edited	Current reviewed value differs from original; an edit alone does not establish correctness

Scoring uses the original draft. Accepting a route suggestion does not rewrite it. Partial references stay partial. Correctly blank fields are displayed separately from correct stated fields; they must not conceal critical instruction errors. Conservative text equality may reject valid paraphrases. The app does not automatically equate every 12 hours with twice daily.

Prepared examples are excluded from archive quality counters. Human role plays, consented recordings and pasted sources remain labelled. Their results must not be presented as an independently sampled clinical benchmark. Speech recognition, extraction, translation and pronunciation require separate assessments.

8 Success metrics and evidence

Metric	Current evidence or definition	Status
Software checks	32 passing checks using fictional fixtures and mocked provider responses	Verified development run for 0.3.1 web
Type and production build checks	TypeScript and deployable build complete successfully	Development verification
Public access	Anonymous visitor can open the page and local demo; private API reads reject missing identity	Anonymous checks passed; account tests pending
Archive outcomes	Counts of correct, incorrect or missed, correctly blank and unsupported fields against verified references	Implemented; no real recording performance claim
Complete case accuracy	Fully correct, fully annotated cases divided by fully annotated cases, accounting for missed and extra medicines	Benchmark reporting remains future work
Review burden	Number of edits and reviewer time before approval	Baseline and study pending
Patient comprehension	Correct source grounded teach back responses after listening	Patient study pending
Audio fidelity	Correctly spoken critical instructions compared with approved facts	Bilingual assessment pending
Latency and cost	Actual stage timings and provider usage, including retries and review	Per run metadata exists; no aggregate latency or rupee claim

The previous Python app's 53 checks are separate evidence and are not added to the website's test count. Passing software checks do not establish medication accuracy, clinical safety or benefit.

9 Manual acceptance checks

Record browser, date, application version, actual result and evidence. Expected behaviour is not a recorded pass. Two account isolation, live API processing and audio quality need their own verification.

Check	Expected result
Open signed out	Public page and research load without owner permission
Open two public examples	Prepared scenarios vary; no key or login is required
Edit public example and refresh	Original remains separate; local changes disappear; no archive write
Request archive signed out	API rejects access; no consultation data appears
Sign in with another account	Its own archive appears; it cannot read the owner's cases or files
Enter and test key	Entry is unverified; an explicit test reports the actual service result
Upload supported 90 second WAV	Batch STT stage and extraction outcome are distinguishable
Clear active session	File, text and current case clear; archived cases remain
Save correction then reopen	Original value, correction and saved event remain available
Annotate absent versus missed fact	Correctly blank, missed and unsupported outcomes stay distinct

Check	Expected result
Confirm route suggestion	It is usable after review but stays excluded from extracted spoken facts
Change approved facts or language	Prior dependent approvals and outputs are invalidated
Edit script and regenerate	Back translation and bilingual review are required again
Listen and export	Reviewed WAV and case JSON agree; no WhatsApp sending claim
Use a stale second tab	A newer saved revision cannot be silently overwritten
Delete own test case	Website case data and archived files are removed; provider records are separate

A separate manual log in the package records release observations and keeps unexecuted tests marked not run. No real recording or patient outcome result is fabricated.

10 Access privacy and GitHub publication

Surface	Access and storage rule
Public website	Fictional preview, research and explanatory material
Uploaded sources and archives	Sign in required; server queries include the authenticated owner ID
Recordings and generated audio	Private file storage, served through an owner checked API
API key	Kept in page memory and sent for model requests; not saved in the archive or browser storage
Retention	Website records remain until case deletion; no automatic retention schedule
Public GitHub repository	Source, PRD, fictional examples and sanitized software evidence; no live records or keys

GitHub repository visibility and website access are independent. The README and repository Website field link to the existing hosted app. Uploading the source does not deploy a second working application. GitHub Pages is static hosting; this app also needs a server, authentication, database and private file storage. GitHub edits do not automatically update the current hosted deployment. GitHub Pages documentation

Recording permission must cover external AI processing and archiving, rather than assuming consultation consent covers those purposes. No production privacy certification, India only residency guarantee, authenticated clinician authorization or legal compliance claim is made.

11 Priorities and next validation

The portfolio release prioritizes an accessible public demonstration, private review records, traceable corrections and honest evaluation. Next, complete a signed in role play from WAV to reviewed audio, verify isolation with two accounts, collect qualified reference annotations and assess bilingual output. Define a held out benchmark before claiming real recording accuracy.

Later work may add authenticated clinician sign off, durable operational controls, measured patient comprehension, appropriate WhatsApp delivery and streaming transcription. Live interpretation requires a separate product and evaluation design; an incomplete streaming transcript should not trigger automatic final instructions.

12 Evidence register

The package contains source code, 32 software checks, test logs, this PRD, market sources and upload instructions. Public files exclude patient records and credentials.